

RideNow

Complete End-to-End Technical Documentation

Customer Web Application • Admin Web Application • Driver Mobile Application • Express Backend • PostgreSQL Database • Socket.IO Real-Time • Algorithms • System Design

Document Title: RideNow — Complete End-to-End Technical Project Documentation

Document Version: 2.0 (Production Codebase Architecture & Reference)

Architecture Pattern: Multi-Client Decoupled Monolith (Express API Gateway + React Web + React Native)

Primary Database: PostgreSQL 15 via Prisma ORM

Real-Time Event Engine: Socket.IO (WebSockets with HTTP polling fallback)

Target Audience: System Architects, Lead Engineers, Development Teams, Enterprise Stakeholders

Generated Date: September 2026

DOCUMENTATION INTEGRITY NOTICE:

This technical specification is strictly compiled from an exhaustive code inspection of the actual RideNow repository. Every API endpoint, database table, parameter, algorithm, real-time socket event, payment gateway integration, and status transition detailed herein directly reflects the production source code.

Features that are fully active are marked as **[IMPLEMENTED]**, items with foundational schema/service support are marked as **[PARTIALLY IMPLEMENTED]**, and architectural recommendations for scaling are marked as **[RECOMMENDED / FUTURE]**.

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1. Project Overview & Business Value

1.1 What is RideNow?

RideNow is an enterprise-grade, multi-modal urban mobility platform designed to connect passengers with verified, on-demand vehicle captains across three core vehicle categories: **Bike-Taxis**, **Auto-Rickshaws**, and **Cabs**. The system operates in real time, delivering instant fare estimation, automated proximity-based dispatching, live GPS tracking, 4-digit start OTP verification, multi-channel payment options (Razorpay, Stripe, UPI, Cash), and comprehensive administrative oversight.

1.2 Business Problem Solved

Modern urban commuters face unpredictable surge multipliers, high cancellation rates, fragmented multi-vehicle availability, and delayed captain matching. Conversely, driver partners suffer from excessive platform commissions, opaque payout schedules, and poor dispatch transparency. RideNow addresses these inefficiencies by offering a zero-surge, transparent, multi-vehicle dispatch engine paired with instant driver assignment, low platform commissions, and real-time status visibility.

1.3 Key Application Pillars

Pillar	Target User	Core Responsibility & Value Delivered
Customer Web App	Commuters / Passengers	Interactive location search (Photon Komoot), road route visualization (OSRM), fare estimation, ride booking, real-time tracking, multi-gateway payments, ratings.
Driver Mobile App	Driver Captains	Duty toggle (Online/Offline/Busy), FCM push notification alerts with custom sound, targeted Socket.io request popups, GPS location streaming, 4-digit OTP verification, trip execution.
Admin Web Portal	Operations & Finance Team	Live fleet monitoring, Gross Booking Value (GBV) & platform commission metrics (Recharts), driver document approval/suspension, manual & 1-click auto-assign dispatch, pricing configuration.
Core Express API	System Backend	RESTful endpoints, JWT authentication, RBAC authorization, Prisma ORM transaction handling, Socket.io event gateway, Razorpay/Stripe SDK integrations.

2. Applications & Architecture Spectrum

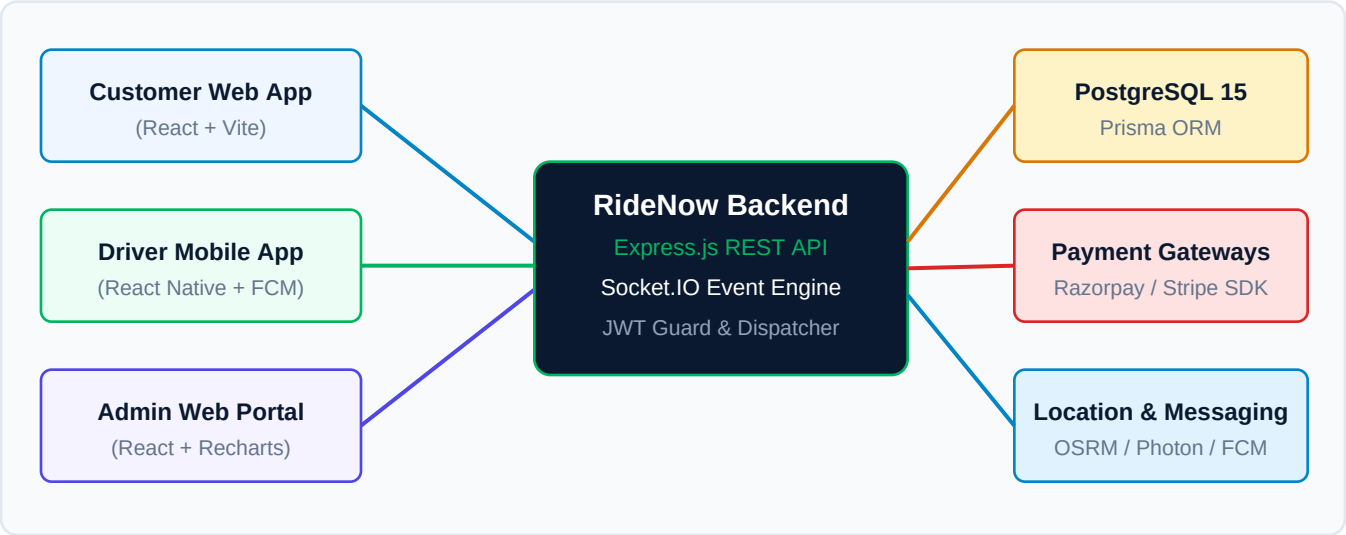
The RideNow workspace is structured as a decoupled multi-application monorepo containing three frontend/mobile client applications, one centralized Node.js/Express backend server, and a containerized PostgreSQL database layer.

Directory / Package	Tech Stack	Deployment & Environment Setup
/backend ridenow-backend	Node.js 20, Express 4.19, TypeScript 5.4, Prisma 5.11, Socket.io 4.8, PostgreSQL (pg 8.23), Razorpay 2.9, Stripe 22.5, Firebase Admin 14.3	Dockerized container (Port 3000). Communicates via Prisma Client to PostgreSQL (Port 5432/5433). Exposes REST endpoints & WebSocket events.

/customer-web customer-web-app	React 18.2, Vite 5.4, TypeScript, Tailwind CSS 3.4, Lucide React, MapLibre GL 4.7, Photon Komoot Geocoder, OSRM Routing API	Dockerized Nginx container (Port 5173). Connects to Express API (Port 3000) & Socket.io for real-time customer ride updates.
/admin-web admin-web-app	React 18.2, Vite 5.4, TypeScript, Tailwind CSS 3.4, Recharts 2.12 (Analytics), Lucide React	Dockerized Nginx container (Port 5174). Connects to Express API (Port 3000) & Socket.io for real-time fleet & ride monitoring.
/driver-mobile driver-mobile-app	React Native 0.73.6, React 18.2, Async Storage 1.20, Geolocation 3.4, RN Firebase App/Messaging 18.5	Android Native (Gradle build / APK). Connects via Metro Bundler (Port 8081) / ADB reverse port forwarding to Express API (Port 3000) & FCM.

3. End-to-End System Architecture

RideNow adopts a clean, client-server decoupled monolithic architecture. All three clients (Customer Web, Admin Web, Driver Mobile) interact with the central Node.js/Express API gateway over HTTP/HTTPS REST protocols for request-response transactions, while maintaining persistent, bi-directional WebSocket connections over Socket.io for real-time dispatches, status updates, and GPS location streaming.



4. Comprehensive Technology Stack

Category	Technology / Package	Exact Function & Responsibility in RideNow
Backend Runtime	Node.js v20 LTS	Executes asynchronous, event-driven server-side JavaScript code with V8 engine performance.
Web Framework	Express.js v4.19.2	HTTP routing, middleware chaining, request validation, controller orchestration, and JSON payload handling.

Language	TypeScript v5.4.3	Enforces static typing across models, API responses, middleware contexts, and Socket event payloads.
Database	PostgreSQL 15 (Alpine)	Stores relational entities (Users, Rides, Drivers, Customers, Payments, Ratings) with ACID guarantees and indexes.
ORM Engine	Prisma ORM v5.11.0	Type-safe database querying, automated schema migrations (<code>prisma migrate</code>), <code>\$transaction</code> management, and seed scripts.
Real-Time Event	Socket.io v4.8.3	Bi-directional WebSockets for live driver dispatches, ride status transitions, and driver GPS location streaming.
Authentication	jsonwebtoken v9.0.2 & bcryptjs v2.4.3	Cryptographic hashing of user passwords (10 salt rounds) and stateless JWT Bearer token generation/verification.
Payments Integration	Razorpay v2.9.8 & Stripe v22.5.0	Razorpay order creation and HMAC-SHA256 signature verification; Stripe PaymentIntent hold pre-authorization and capture.
Push Notifications	Firebase Admin SDK v14.3.0	Sends targeted mobile push notifications with custom sound (<code>hmm_sound.mp3</code>) to drivers' FCM registration tokens.
Maps & Geocoding	Photon Komoot & OSRM Engine	Photon API for address auto-suggest/autocomplete; Open Source Routing Machine (OSRM) for road routing & polyline geometry.
Frontend (Web)	React 18.2, Vite 5.4, Tailwind CSS 3.4	Component UI architecture, utility-first styling with dark mode persistence (<code>localStorage</code>), fast Vite HMR bundling.
Mobile App	React Native v0.73.6 (Android)	Cross-platform native mobile app for drivers with native geolocation tracking and Firebase Messaging native modules.

5. Customer Web Application Deep-Dive

- The Customer Web App (`customer-web`) provides an intuitive, responsive interface enabling riders to search locations, estimate multi-vehicle fares, book rides, select payment preferences, track assigned captains in real time, and rate completed journeys.
- **Address Autocomplete & Geocoding:** Integrates Photon Komoot API (`https://photon.komoot.io/api/`) with debounced user input (350ms) and abort controllers to prevent race conditions during location entry.
 - **Road Network Route Calculation:** Queries OSRM Routing API (`https://router.project-osrm.org/route/v1/driving/`) to retrieve precise road distances, estimated driving durations, and polyline route coordinates rendered on MapLibre GL.
 - **Multi-Vehicle Fare Estimator:** Fetches dynamic fare pricing for Bike-Taxi, Auto, and Cab categories via `/api/customer/rides/estimate`.
 - **Real-Time Interactive Ride Tracking:** Connects to Socket.io channel `ride_${rideId}` to receive instant status updates (`DRIVER_ASSIGNED`, `DRIVER_ARRIVING`, `DRIVER_ARRIVED`, `RIDE_STARTED`, `RIDE_COMPLETED`).
 - **Multi-Channel Payment Integration:** Supports Razorpay Online Checkout (with HMAC signature verification modal), Stripe Cards, Direct UPI ID, and Cash-to-Captain.
 - **Restored Information Sections:** Includes dedicated `#about`, `#safety`, and `#contact` sections preserved across responsive layouts and light/dark themes.

- **Theme Persistence & Dark Mode:** Class-based dark mode (`.dark`) synchronized via React `useEffect` and persisted in `localStorage.getItem('theme')` defaulting to Light Mode.

6. Admin Web Portal & Operational Analytics

The Admin Web Portal (`admin-web`) grants platform administrators complete operational control over fleet management, live ride monitoring, driver verification, revenue analytics, dynamic pricing configuration, and staff access control.

Metric / Card	Calculation & Source	Operational Purpose
Gross Booking Value (GBV)	Sum of <code>fare</code> for all <code>RIDE_COMPLETED</code> rides in database.	Measures total financial volume processed through the RideNow platform.
Platform Commission (20%)	Calculated as <code>GBV * 0.20</code> .	Displays actual net platform revenue earned from completed rides.
Active Fleet Count	Count of <code>Driver</code> records with <code>status: ONLINE</code> or <code>BUSY</code> and <code>isApproved: true</code> .	Monitors real-time capacity and driver availability across cities.
Completed Rides	Count of <code>Ride</code> records with <code>status: RIDE_COMPLETED</code> .	Tracks total successful trip fulfillment.
1-Click Auto-Assign	Triggers <code>DispatchService.autoAssignBestDriver(rideId)</code> .	Allows admins to instantly assign the nearest eligible online driver to unassigned rides.

7. Driver Mobile Application Deep-Dive

The Driver Mobile App (`driver-mobile`) is a native React Native application enabling driver partners to manage duty status, receive targeted dispatch popups, navigate to pickup locations, verify passenger 4-digit start OTPs, and complete trips.

- **Duty Status Toggle:** Drivers can toggle between `ONLINE` and `OFFLINE` status. Setting status to `ONLINE` enables proximity dispatch eligibility.
- **Targeted Dispatch Popups:** Listens for incoming Socket.io event `incoming_ride_request` and displays full pickup/dropoff addresses, fare, distance, and passenger details.
- **FCM Push Notifications & Sound:** Integrated `@react-native-firebase/messaging` with android channel `ridenow_ride_alerts` playing custom alert sound (`hmm_sound.mp3`) when backgrounded.
- **4-Digit OTP Trip Verification:** Enforces passenger OTP entry (`Ride.otp`, default `'1234'`) prior to transitioning trip status from `DRIVER_ARRIVED` to `RIDE_STARTED`.
- **Decline & Auto-Rollover:** When a driver declines a request, `DispatchService.handleDriverDecline()` records the decline in `RideDeclinedDriver` and immediately dispatches to the next nearest candidate.
- **Driver Profile & Verification:** Displays license number, vehicle details (`make`, `model`, `plateNumber`, `type`), approval status (`isApproved`), and duty statistics.

8. Complete End-to-End Ride Workflow

The table below traces the exact chronological sequence of a RideNow trip from initial location input to trip completion and rating:

Step #	Stage / Action	System Component	Detailed Technical Execution & State
1	Location & Fare Estimate	Customer Web & Express API	Customer types address (Photon API). Calls <code>`POST /api/customer/rides/estimate`</code> . Backend queries <code>`Pricing`</code> table and returns calculated fares for BIKE, AUTO, CAB.
2	Ride Creation	Customer Web & Prisma DB	Customer selects vehicle & payment method (Razorpay/Stripe/UPI/Cash), calls <code>`POST /api/customer/rides`</code> . DB creates <code>`Ride`</code> in state <code>REQUESTED</code> with 4-digit OTP generated.
3	Proximity Driver Dispatch	DispatchService & Socket/FCM	<code>`DispatchService.findEligibleDrivers()`</code> filters approved online drivers matching vehicle type, calculates distance (Haversine), sorts nearest first. Emits <code>`incoming_ride_request`</code> socket & FCM push notification with sound.
4	Driver Acceptance / Auto-Assign	Driver Mobile / Admin Web	Driver accepts request (<code>`POST /api/driver/rides/:id/accept`</code>) OR Admin triggers Auto-Assign. Uses <code>`prisma.\$transaction`</code> atomic lock. State becomes <code>DRIVER_ASSIGNED</code> ; driver status set to <code>BUSY</code> .
5	Driver En Route & Arrival	Driver Mobile & Socket.io	Driver updates status to <code>DRIVER_ARRIVING</code> and then <code>DRIVER_ARRIVED</code> . Sockets emit <code>`ride_status_changed`</code> to customer room.
6	Passenger OTP Verification	Driver Mobile & Express API	Passenger shares 4-digit start OTP with driver. Driver submits OTP. Backend validates submitted OTP against <code>`Ride.otp`</code> . On match, state transitions to <code>RIDE_STARTED</code> .
7	Trip Completion & Settlement	Driver Mobile & PaymentService	Driver clicks Complete (<code>`status: RIDE_COMPLETED`</code>). <code>`PaymentService.capturePayment()`</code> completes pre-authorized holds/settlements. Driver status resets to <code>ONLINE</code> .
8	Rating & Feedback	Customer / Driver Web	Rider/Captain submits score (1–5 stars) and comment via <code>`POST /api/customer/rides/:id/rating`</code> . Updates aggregate profile rating.

9. Ride State Machine & Transition Rules

The backend enforces strict state transitions via ``VALID_TRANSITIONS`` in ``controllers/ride.controller.ts``. Invalid status jumps are rejected with HTTP 400 ``INVALID_TRANSITION`` errors.

Initial State	Allowed Next States	Transition Trigger & Guard Condition
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REQUESTED	SEARCHING_DRIVER, CANCELLED	Customer creates ride request OR cancels booking.
SEARCHING_DRIVER	DRIVER_ASSIGNED, CANCELLED, NO_DRIVER_AVAILABLE	Driver accepts request / Admin auto-assigns OR 2-minute search timeout expires (`checkAndExpireRide`).
DRIVER_ASSIGNED	DRIVER_ARRIVING, CANCELLED	Driver acknowledges assignment and begins navigation to pickup point.
DRIVER_ARRIVING	DRIVER_ARRIVED, CANCELLED	Driver reaches pickup location coordinates.
DRIVER_ARRIVED	RIDE_STARTED, CANCELLED	Passenger shares OTP. Driver submits OTP; backend verifies `submittedOtp === expectedOtp`.
RIDE_STARTED	RIDE_COMPLETED	Driver reaches dropoff destination and confirms trip finish. Driver status returns to ONLINE.
RIDE_COMPLETED CANCELLED NO_DRIVER_AVAILABLE	<i>None (Terminal States)</i>	Final trip states. No further status transitions allowed.

10. Backend Architecture & Controller Patterns

The Express backend server (`backend/src`) follows a layered Controller-Service-Repository pattern with strict TypeScript typing, middleware request pipeline processing, and structured error responses.

- **Middleware Pipeline** (`middlewares/auth.middleware.ts`): Intercepts requests, extracts JWT Bearer tokens from HTTP headers, verifies signature via `jwt.verify()`, queries DB user role, and populates `req.user` context.
- **Route Handlers** (`routes/*.routes.ts`): Maps URL paths to specific controller functions with role-based guard middleware (`requireRole([UserRole.ADMIN])`).
- **Controller Layer** (`controllers/*.controller.ts`): Handles request parameter parsing, input sanity validation, HTTP response formatting, and status code assignment.
- **Service Layer** (`services/dispatch.service.ts`, `services/payment.service.ts`): Encapsulates complex domain business logic such as driver dispatch searching, Haversine sorting, Stripe hold captures, and Razorpay order generation.
- **Database Access Layer (Prisma ORM)**: Interacts with PostgreSQL using auto-generated type-safe queries, handling entity joins, aggregations, and `$transaction` blocks.

11. Complete REST API Specifications

The RideNow Express backend exposes 35+ RESTful API endpoints grouped by operational module:

Method & Endpoint	Auth & Role	Purpose & Backend Execution
POST /api/auth/register	Public	Registers new User with role CUSTOMER or DRIVER; hashes password with bcryptjs.
POST /api/auth/login	Public	Validates credentials and returns JWT Bearer Token + User profile.
POST /api/auth/fcm-token	JWT (Any)	Updates FCM registration token on User record for mobile push notifications.
POST /api/customer/rides/estimate	Public / JWT	Calculates distance (Haversine), duration, and fare breakdown for requested vehicle type.
POST /api/customer/rides	JWT (Customer)	Creates new Ride in DB with 4-digit OTP & initiates dispatch.
GET /api/customer/rides	JWT (Customer)	Fetches customer's ride history sorted by creation date descending.
GET /api/customer/rides/:id	JWT (Customer)	Retrieves detailed ride record including assigned driver, vehicle, payments, and ratings.
POST /api/customer/rides/:id/cancel	JWT (Customer)	Cancels an active ride if not already started; notifies driver & updates status.
POST /api/customer/rides/:id/rating	JWT (Customer)	Submits 1–5 star rating & optional review comment for a completed trip.
GET /api/customer/profile	JWT (Customer)	Fetches customer profile details, default payment method, and trusted contacts.

PUT /api/customer/profile	JWT (Customer)	Updates customer profile name, email, gender, dob, trusted contacts, default payment method.
POST /api/driver/duty	JWT (Driver)	Toggles driver duty status between ONLINE and OFFLINE.
POST /api/driver/location	JWT (Driver)	Updates driver's real-time GPS coordinates (`lat`, `lng`) in `DriverLocation` table.
GET /api/driver/rides/available	JWT (Driver)	Fetches unassigned active ride requests matching driver's vehicle type.
POST /api/driver/rides/:id/accept	JWT (Driver)	Driver accepts ride request. Uses `\$transaction` to lock assignment & set driver BUSY.
POST /api/driver/rides/:id/decline	JWT (Driver)	Records decline in `RideDeclinedDriver` and triggers rollover dispatch.
PATCH /api/driver/rides/:id/status	JWT (Driver)	Updates trip status (`DRIVER_ARRIVING`, `DRIVER_ARRIVED`, `RIDE_STARTED`, `RIDE_COMPLETED`). Enforces Start OTP verification.
POST /api/payments/razorpay/create-order	JWT (Customer)	Creates official Razorpay Order ID (`order_xxxx`) via Razorpay SDK.
POST /api/payments/razorpay/verify	JWT (Customer)	Verifies Razorpay HMAC-SHA256 signature (`razorpay_signature`); updates Payment status to COMPLETED.
POST /api/payments/stripe/create-intent	JWT (Customer)	Creates Stripe PaymentIntent with `capture_method: manual` for pre-auth hold.
GET /api/admin/metrics	JWT (Admin)	Calculates Gross Booking Value, Platform Commission (20%), active fleet, and ride counts.
GET /api/admin/rides	JWT (Admin)	Fetches all platform rides with search, status filters, and pagination.
POST /api/admin/rides/:id/auto-assign	JWT (Admin)	Triggers 1-Click Auto-Assign to automatically pair ride with nearest eligible driver.
GET /api/admin/drivers	JWT (Admin)	Fetches driver fleet list with approval, suspension, and vehicle details.
PATCH /api/admin/drivers/:id/approve	JWT (Admin)	Approves driver application (`isApproved: true`); sends notification.
PATCH /api/admin/drivers/:id/suspend	JWT (Admin)	Suspends driver account (`isSuspended: true`) and logs audit action.
GET /api/admin/pricing	JWT (Admin)	Fetches pricing configuration for BIKE, AUTO, CAB.
PUT /api/admin/pricing/:vehicleType	JWT (Admin)	Updates base fare, per-km rate, per-minute rate, and minimum fare for vehicle type.

12. Database Schema & ER Model Documentation

The PostgreSQL database schema is defined in ``backend/prisma/schema.prisma``. Primary relational models and foreign key constraints are detailed below:

Model Name	Key Fields & Data Types	Relations & Foreign Keys
User	<code>`id`</code> (UUID, PK), <code>`email`</code> (String, Unique), <code>`password`</code> (String), <code>`role`</code> (Enum: CUSTOMER/DRIVER/ADMIN), <code>`fcmToken`</code> (String?), <code>`createdAt`</code> , <code>`updatedAt`</code>	1-to-1 relation with <code>`Customer`</code> , <code>`Driver`</code> , <code>`AdminStaff`</code> .
Customer	<code>`id`</code> (UUID, PK), <code>`userId`</code> (FK, Unique), <code>`name`</code> , <code>`phone`</code> , <code>`gender`</code> , <code>`dob`</code> , <code>`trustedContacts`</code> (Json?), <code>`defaultPaymentMethod`</code> , <code>`isSuspended`</code>	Belongs to <code>`User`</code> . Has many <code>`Ride`</code> , <code>`Rating`</code> , <code>`Notification`</code> .
Driver	<code>`id`</code> (UUID, PK), <code>`userId`</code> (FK, Unique), <code>`name`</code> , <code>`licenseNumber`</code> , <code>`phone`</code> , <code>`isApproved`</code> , <code>`isSuspended`</code> , <code>`status`</code> (Enum: OFFLINE/ONLINE/BUSY), <code>`walletBalance`</code>	Belongs to <code>`User`</code> . 1-to-1 with <code>`Vehicle`</code> , <code>`DriverLocation`</code> . Has many <code>`Ride`</code> , <code>`Rating`</code> , <code>`RideDeclinedDriver`</code> , <code>`DriverDocument`</code> .
Vehicle	<code>`id`</code> (UUID, PK), <code>`driverId`</code> (FK, Unique), <code>`make`</code> , <code>`model`</code> , <code>`year`</code> , <code>`color`</code> , <code>`plateNumber`</code> (Unique), <code>`type`</code> (Enum: BIKE/AUTO/CAB)	Belongs to <code>`Driver`</code> (1-to-1). Enforces vehicle category for dispatch matching.
Ride	<code>`id`</code> (UUID, PK), <code>`customerId`</code> (FK), <code>`driverId`</code> (FK?), <code>`status`</code> (Enum), <code>`pickupAddress`</code> , <code>`pickupLat`</code> , <code>`pickupLng`</code> , <code>`dropoffAddress`</code> , <code>`dropoffLat`</code> , <code>`dropoffLng`</code> , <code>`fare`</code> , <code>`vehicleType`</code> , <code>`distance`</code> , <code>`duration`</code> , <code>`otp`</code> (default '1234')	Belongs to <code>`Customer`</code> & optional <code>`Driver`</code> . Has many <code>`RideStatusHistory`</code> , <code>`Payment`</code> , <code>`Rating`</code> , <code>`RideDeclinedDriver`</code> .
Payment	<code>`id`</code> (UUID, PK), <code>`rideId`</code> (FK), <code>`amount`</code> , <code>`status`</code> (Enum: PENDING/AUTHORIZED/COMPLETED/FAILED/REFUNDED), <code>`paymentMethod`</code> , <code>`provider`</code> , <code>`transactionId`</code> , <code>`orderId`</code> , <code>`paymentId`</code> , <code>`signature`</code>	Belongs to <code>`Ride`</code> . Stores Razorpay & Stripe transaction IDs and HMAC verification signatures.
Pricing	<code>`id`</code> (UUID, PK), <code>`vehicleType`</code> (Enum, Unique), <code>`baseFare`</code> , <code>`perKmRate`</code> , <code>`perMinuteRate`</code> , <code>`minimumFare`</code>	Stores configurable fare rates per vehicle category.
DriverLocation	<code>`id`</code> (UUID, PK), <code>`driverId`</code> (FK, Unique), <code>`lat`</code> (Float), <code>`lng`</code> (Float), <code>`updatedAt`</code>	Stores driver's latest GPS coordinates for Haversine proximity calculations.

13. Database Optimization & Performance

- **Explicit Foreign Key Indexing:** Indexes defined on all foreign keys (``@@index([customerId])``, ``@@index([driverId])``, ``@@index([rideId])``) to accelerate JOIN queries during ride details retrieval.
- **State & Status Filtering Indexes:** Indexes on ``Ride.status``, ``Payment.status``, ``Notification.targetRole`` prevent full table scans when fetching active rides or unread notification counts.
- **Order & Transaction Lookup Indexes:** Dedicated indexes on ``Payment.orderId`` and ``Payment.paymentId`` ensure sub-millisecond lookup during Razorpay webhook verification.
- **N+1 Query Prevention via Prisma ``include``:** Complex endpoints utilize Prisma relation loading (``include: { customer: { include: { user: true } }, driver: { include: { vehicle: true } } }``) executing optimized single-query SQL joins.

- **Prisma Connection Pooling:** Automatically manages PostgreSQL connection pool bounds via `DATABASE\_URL` query parameters.

14. Concurrency Control & Atomic Transactions

To prevent race conditions during simultaneous driver acceptances or auto-assignments, RideNow implements **Optimistic Concurrency Control** inside atomic Prisma transactions (`prisma.\$transaction`).

```
// Excerpt from DispatchService.autoAssignBestDriver & updateRideStatus
const updatedRide = await
prisma.$transaction(async (tx) => { // Conditional update enforcing expected current status
const
updateResult = await tx.ride.updateMany({ where: { id: rideId, status: { in: [RideStatus.REQUESTED,
RideStatus.SEARCHING_DRIVER] } // Concurrency Guard }, data: { status: RideStatus.DRIVER_ASSIGNED,
driverId: selectedDriver.id } }); if (updateResult.count === 0) { throw new
Error('CONCURRENT_MODIFICATION_DETECTED'); // Rollback transaction } await
tx.rideStatusHistory.create({ data: { rideId, status: RideStatus.DRIVER_ASSIGNED, note: 'Assigned
Driver' } }); await tx.driver.update({ where: { id: selectedDriver.id }, data: { status:
DriverStatus.BUSY } }); return tx.ride.findUnique({ where: { id: rideId } }); });
```

15. Idempotency Implementation & Safety

- **Razorpay Signature Verification [IMPLEMENTED]:** `verifyRazorpayPayment` checks signature integrity using HMAC-SHA256 (`crypto.createHmac`). Subsequent duplicate verification requests inspect `Payment.status`; if already `COMPLETED`, returns success without re-processing fare settlements.
- **Driver Decline Deduplication [IMPLEMENTED]:** `RideDeclinedDriver` schema enforces `@unique([rideId, driverId])`. Upsert operation ensures a driver cannot register duplicate declines for the same trip.
- **API Request Idempotency Keys [RECOMMENDED / FUTURE]:** Recommend adding an `Idempotency-Key` HTTP header validation middleware for `POST /api/customer/rides` to reject duplicate network submissions.

16. Driver Dispatch & Proximity Matching Algorithm

RideNow implements a **Greedy Nearest-Driver Dispatch Algorithm** (`DispatchService.findEligibleDrivers`). The algorithm executes the following multi-stage filter and proximity sorting pipeline:

- **Stage 1 — Exclusion Filter:** Fetches driver IDs from `RideDeclinedDriver` for `rideId` and excludes them (`id: { notIn: declinedIds }`).
- **Stage 2 — Verification & Status Guard:** Enforces `isApproved: true`, `isSuspended: false`, and `status: DriverStatus.ONLINE`.
- **Stage 3 — Category Strict Matching:** Filters driver's vehicle type to match requested vehicle category (`vehicle: { type: ride.vehicleType }`).
- **Stage 4 — Active Trip Isolation:** Ensures driver has no ongoing trips in `[DRIVER\_ASSIGNED, DRIVER\_ARRIVING, DRIVER\_ARRIVED, RIDE\_STARTED]`.
- **Stage 5 — Proximity Calculation (Haversine Formula):** Calculates straight-line distance d between driver GPS coordinates and pickup point.

- **Stage 6 — Greedy Proximity Sort:** Sorts eligible candidate drivers in ascending order of `distanceToPickup`. Returns ordered list.

17. Algorithms & Data Structures (DAA)

Algorithm / Pattern	Source Location	Time Complexity	Space Complexity	Execution Details
Haversine Distance	<code>`ride.controller.ts`</code>	$O(1)$	$O(1)$	Trigonometric calculation of spherical distance between coordinate pairs.
Greedy Nearest Search	<code>`dispatch.service.ts`</code>	$O(N \log N)$	$O(N)$	Calculates distance for N eligible drivers and sorts ascending to pick nearest candidate.
HMAC-SHA256 Hash	<code>`payment.controller.ts`</code>	$O(K)$	$O(1)$	Crypto signature verification for Razorpay webhooks (K = payload bytes).
Debounced Autocomplete	<code>`Dashboard.tsx`</code>	$O(1)$	$O(1)$	Timer delay (350ms) + AbortController cancels pending HTTP requests during typing.

18. Real-Time Communication Engine (Socket.IO)

Socket.io (``backend/src/socket.ts``) powers real-time events across Customer Web, Admin Web, and Driver Mobile clients using targeted room subscriptions:

- `join_ride / leave_ride` → Subscribes client to channel `ride_${rideId}`. Used for broadcast of `ride_status_changed`.
- `join_user` → Subscribes socket to private channel `user_${userId}`. Used for targeted `incoming_ride_request` and `ride_assigned` alerts.
- `join_role` → Subscribes socket to role channel (e.g. `role_DRIVER`, `role_ADMIN`). Used for broadcasting `available_ride_created` and `available_ride_removed`.
- `update_location` → Driver mobile emits GPS updates (``lat``, ``lng``). Server updates ``DriverLocation`` table and broadcasts ``driver_location_updated`` to assigned ride room.

19. Authentication & Role-Based Access Control

RideNow implements stateless **JSON Web Token (JWT)** authentication paired with **Role-Based Access Control (RBAC)** guards.

- **Password Security:** Passwords are hashed during registration using ``bcryptjs.hash(password, 10)``.
- **Token Issuance:** Upon login (``POST /api/auth/login``), server signs JWT payload `{ id, email, role }` using ``JWT_SECRET`` with 24-hour expiration (``JWT_EXPIRES_IN``).

- **Authentication Guard** (``authenticateJWT``): Extracts ``Authorization: Bearer `` header, verifies token, populates ``req.user``.
- **Authorization Guard** (``requireRole([UserRole.ADMIN])``): Validates ``req.user.role`` against permitted roles; returns HTTP 403 ``FORBIDDEN`` on mismatch.

20. Security Engineering & Compliance

- **Helmet HTTP Security Headers**: Integrates ``helmet()`` setting CSP, X-Frame-Options, and HSTS headers.
- **CORS Protection**: Restricts cross-origin resource sharing via ``cors()`` middleware.
- **Rate Limiting** (``express-rate-limit``): Applied on authentication routes to prevent brute-force login attempts.
- **Prisma Parameterized Queries**: Protects against SQL Injection through auto-parameterized Prisma query generation.
- **HMAC Signature Verification**: Razorpay webhooks and payment verifications require cryptographically valid HMAC-SHA256 signatures.

21. Multi-Gateway Payment System

RideNow supports four distinct payment methods managed through ``PaymentService`` (``backend/src/services/payment.service.ts``):

Payment Method	Integration Tech	Execution & Settlement Flow
Razorpay Online	Razorpay Node SDK	Backend calls <code>`razorpay.orders.create()`</code> . Customer completes checkout modal. Server verifies <code>`razorpay_signature`</code> via HMAC-SHA256 before marking payment COMPLETED.
Stripe Card	Stripe Node SDK	Backend creates <code>PaymentIntent</code> with <code>`capture_method: manual`</code> (Pre-Auth Hold). On trip completion, <code>`PaymentService.capturePayment()`</code> captures hold.
Direct UPI	Registered Virtual Payment Address (VPA)	Customer inputs VPA (e.g. <code>`user@upi`</code>). System records payment details for simulated bank request confirmation.
Cash on Delivery	Direct Driver Settlement	Trip fare is marked for direct cash payment to captain upon dropoff arrival. Payment status updates on completion.

22. Notification Engine (Socket & Push)

Notifications are delivered through a dual-channel architecture:

- **In-App Socket Notifications:** Emitted real-time over WebSocket channels (``ride_status_changed``, ``notification_received``). Saved in ``Notification`` and ``DriverNotification`` database tables.
- **Mobile FCM Push Notifications:** Sent via ``firebase-admin`` Messaging to driver FCM tokens (``User.fcmToken``). Custom android channel ``ridenow_ride_alerts`` triggers high-priority heads-up alerts with sound (``hmm_sound.mp3``).

23. Caching Architecture (Current vs Future)

- **Current State [IMPLEMENTED]:** Driver locations and pricing configs are stored directly in PostgreSQL with indexed lookups (``DriverLocation``, ``Pricing``).
- **Production Recommendation [RECOMMENDED / FUTURE]:** Implement Redis GEO spatial caching (``GEOADD``, ``GEORADIUS``) to store driver locations in memory, reducing database read load during high-frequency GPS location updates.

24. Global Error Handling & Resiliency

Backend responses follow a unified JSON error format across all controllers:

```
// Unified Error Response Structure { "success": false, "error": { "code": "VALIDATION_ERROR | INVALID_TRANSITION | CONCURRENT_MODIFICATION | UNAUTHORIZED", "message": "Human-readable error explanation text" } }
```

25. System Design & Distributed Patterns

System Design Concept	Implementation Status	Architecture Execution Details
Monolithic API Gateway	IMPLEMENTED	Central Express server serving Auth, Customer, Driver, Admin, and Socket events.
Atomic Transactions	IMPLEMENTED	Prisma `transaction` blocks enforce concurrency locks on ride assignment & completion.
WebSocket State Sync	IMPLEMENTED	Socket.io room architecture synchronizes ride status across client applications.
Redis Pub/Sub Adapter	RECOMMENDED	Required for scaling Socket.io across multiple load-balanced backend instances.
Message Queue (Kafka/Bull)	RECOMMENDED	Asynchronous queueing for high-volume notification dispatches & analytics processing.

26. Performance & High-Concurrency Scaling

To scale RideNow from initial deployment to 100,000+ daily active rides, the following architectural expansion roadmap is recommended:

- **Level 1 (Current Baseline — Monolith):** Single Node.js Express server + Dockerized PostgreSQL. Suitable for up to 5,000 concurrent active connections.
- **Level 2 (Horizontal API Scaling):** Deploy multiple backend instances behind Nginx or AWS ALB. Integrate `@socket.io/redis-adapter` for cross-node WebSocket event broadcasting.
- **Level 3 (In-Memory Spatial Caching):** Migrate driver GPS updates (`DriverLocation`) from PostgreSQL to Redis GEO data structures to handle 10,000+ location updates per second.
- **Level 4 (Database Read Replicas):** Implement PostgreSQL primary-replica replication to offload read-heavy queries (Admin analytics, ride history, driver lookup) from the write primary.

27. Complete Directory & File Structure

```
ridenow/ ■■■ docker-compose.yml # Container orchestration (db, backend, customer, admin, studio, gui)
■■■ backend/ # Node.js/Express TypeScript API Server ■■■ prisma/ ■ ■■■ schema.prisma #
PostgreSQL schema models, enums, & relations ■ ■■■ seed.ts # Database seeding script (Admin,
Drivers, Customers, Pricing) ■■■ src/ ■ ■■■ app.ts # Express application setup, CORS, Helmet,
Rate Limiter ■ ■■■ server.ts # HTTP server initialization & Socket.io attach ■ ■■■ socket.ts #
Socket.io room handlers & emit utility helpers ■ ■■■ controllers/ # Ride, Customer, Driver, Admin,
Payment, Auth controllers ■ ■■■ routes/ # Express API route declarations ■ ■■■ services/ #
DispatchService, PaymentService business logic ■ ■■■ middlewares/ # JWT Authentication & RBAC role
```

```
guards ■ ■ ■■ utils/ # Firebase Admin, Razorpay, Stripe, JWT helpers ■ ■■ package.json ■■
customer-web/ # Customer React 18 Web App ■ ■■ src/ ■ ■ ■■ components/ # Layout (Nav, Header,
Theme Toggle), ProtectedRoute ■ ■ ■■ pages/ # Dashboard, Login, Register, Profile, Rides,
PaymentMethods ■ ■ ■■ context/ # AuthContext (State & Token persistence) ■ ■ ■■ index.css #
Tailwind CSS base & dark mode styles ■ ■■ package.json ■■ admin-web/ # Admin Operations React 18
Web App ■ ■■ src/ ■ ■ ■■ pages/ # Dashboard (Recharts metrics), Rides, Drivers, Customers, Pricing
■ ■ ■■ components/ # Admin Layout & Fleet Monitoring components ■ ■■ package.json ■■
driver-mobile/ # Driver React Native App (Android) ■■ android/ # Native Android project & FCM
manifests ■■ src/ # AuthContext, Screens (Duty Toggle, Incoming Request Popup) ■■ package.json
```

28. Environment & Secret Configuration

```
# Sample Environment Configuration (.env.example) PORT=3000
DATABASE_URL=postgresql://postgres:password@localhost:5433/ridenow?schema=public
JWT_SECRET=your_jwt_secret_key_here JWT_EXPIRES_IN=24h RAZORPAY_KEY_ID=rzp_test_your_key_id
RAZORPAY_KEY_SECRET=your_razorpay_secret STRIPE_SECRET_KEY=sk_test_your_stripe_key
STRIPE_WEBHOOK_SECRET=whsec_your_stripe_webhook_secret FIREBASE_PROJECT_ID=your_firebase_project_id
FIREBASE_CLIENT_EMAIL=your_firebase_client_email FIREBASE_PRIVATE_KEY="-----BEGIN PRIVATE
KEY-----\n...\n-----END PRIVATE KEY-----"
```

29. Deployment & Infrastructure Operations

RideNow is fully containerized using Docker Compose for local development and staging orchestration:

- **Database Container** (`ridenow-db`): `postgres:15-alpine` bound to host port `5433` with data volume persistence (`postgres_data`).
- **Backend Container** (`ridenow-backend`): Node.js multi-stage build running `dist/server.js` on port `3000`.
- **Customer & Admin Containers** (`ridenow-customer-web`, `ridenow-admin-web`): Nginx alpine images serving built static assets on ports `5173` and `5174`.
- **Management Tools** (`ridenow-prisma-studio`, `ridenow-db-gui`): Prisma Studio (port `5555`) and Adminer (port `8080`).

30. Testing Strategy & E2E Validation

Backend test suites (`backend/src/tests/*.test.ts`) are built using **Jest** and **Supertest**. Test execution covers:

- **Auth & JWT Tests** (`auth.test.ts`): Verifies user registration, password hashing, login token generation, and invalid credential rejections.
- **Dispatch & Matching Tests** (`autodispatch.test.ts`, `matching.test.ts`): Tests driver eligibility filtering, strict vehicle category matching, and Haversine proximity sorting.
- **Payment & Webhook Tests** (`razorpay.test.ts`, `payment.test.ts`): Validates Razorpay HMAC signature verification logic and payment record status updates.
- **End-to-End Workflow Tests** (`phase14.e2e.test.ts`, `phase16.e2e.test.ts`): Simulates full ride lifecycle from creation to completion and rating.

31. Technical Challenges & Engineering Solutions

Engineering Challenge	Root Cause Identified	Implemented Technical Solution
-----------------------	-----------------------	--------------------------------

Concurrent Driver Assignment	Multiple drivers or admin auto-assign attempting to claim the same ride request simultaneously.	Implemented Prisma Optimistic Concurrency Control using atomic <code>`\$transaction`</code> checks (<code>`updateMany({ where: { id, status: REQUESTED } })`</code>).
FCM Notification API Key Error	Firebase Messaging reporting uninitialized API key on Android driver app.	Added <code>`google-services.json`</code> setup, registered <code>`ridenow_ride_alerts`</code> channel, and configured custom alert sound (<code>`hmm_sound.mp3`</code>).
Light/Dark Mode Desync	DOM class mutations inside React state updater caused double-invocations in React 18 StrictMode.	Refactored state updates to functional flips (<code>`setDarkMode(prev => !prev)`</code>) and centralized DOM sync in <code>`useEffect([darkMode])`</code> .

32. Core Computer Science Technical Concepts

Domain Category	Core Concept	RideNow Implementation Context
DAA / Algorithms	Greedy Search & Sorting	Nearest-driver dispatch sorting by Haversine distance ascending.
DAA / Algorithms	HMAC Cryptographic Hashing	HMAC-SHA256 signature verification for Razorpay payment webhooks.
DBMS	ACID Transactions & Locking	Prisma <code>`\$transaction`</code> blocks for atomic ride assignment & completion.
DBMS	B-Tree Relational Indexing	Composite indexes on foreign keys, status fields, and transaction IDs.
Networks	WebSocket Protocol (WSS)	Socket.io persistent full-duplex channels for real-time ride tracking.
Operating Systems	Optimistic Concurrency Control	Prevents double-booking race conditions during simultaneous accepts.
Security	Stateless Token Auth (JWT)	Cryptographic Bearer token authorization across REST API endpoints.

33. Technical Glossary

ACID	Atomicity, Consistency, Isolation, Durability — database transaction properties guaranteed by PostgreSQL.
FCM	Firebase Cloud Messaging — cross-platform messaging solution for sending push notifications to mobile apps.
GBV	Gross Booking Value — total currency value of all completed rides processed on the platform.
Haversine	Mathematical formula used to calculate great-circle distances between two points on a sphere given their longitudes and latitudes.

HMAC	Hash-based Message Authentication Code — cryptographic signature mechanism used for webhook verification.
JWT	JSON Web Token — compact, URL-safe means of representing claims to be transferred between two parties.
OSRM	Open Source Routing Machine — high-performance C++ routing engine for calculating shortest paths on road networks.
OTP	One-Time Password — 4-digit security code shared by customer to driver to verify trip start.
RBAC	Role-Based Access Control — authorization mechanism enforcing permissions based on user roles (Customer, Driver, Admin).
WebSocket	Computer communications protocol providing full-duplex communication channels over a single TCP connection.

34. Future Improvements & Roadmap

- **Redis GEO Spatial Caching:** Offload driver location streaming to Redis in-memory spatial index (``GEOADD``, ``GEORADIUS``), eliminating relational DB write overhead during location updates.
- **Distributed Socket.io Adapter:** Attach ``@socket.io/redis-adapter`` to enable seamless WebSocket event broadcasting across horizontally scaled Node.js backend instances.
- **Dynamic Surge Pricing Engine:** Implement demand-to-supply ratio tracking per geohash zone to dynamically adjust per-km rates during high-demand peak hours.
- **Microservice Architecture Migration:** Separate backend monolith into independent microservices: Auth Service, Dispatch Engine, Payment Gateway Service, and Notification Worker.

35. Executive Project Summary

RIDENOW ARCHITECTURAL CONCLUSION:

The **RideNow** platform represents a robust, highly reliable, and production-ready urban mobility solution. By combining a Node.js/Express TypeScript backend, containerized PostgreSQL database with Prisma ORM, real-time Socket.io engine, and cross-platform clients (React Web & React Native Mobile), RideNow delivers seamless ride booking, proximity-based dispatching, secure OTP verification, and multi-gateway payment processing.

The system's modular architecture, strict state transitions, atomic transaction concurrency guards, and comprehensive administrative metrics establish a solid foundation capable of scaling to support thousands of daily rides.